Experiments in Physics: PHYS4350_01 Spring 2023
Boston College, Department of Physics
Syllabus

Scheduled class meeting time:
Tuesday and Thursday: 1:30 – 4:00 PM, Room = 130 & 250 Higgins Hall

Required Texts:
There is no textbook is required to be purchased for this course. Journal articles, books and other materials required for this course and will be made available to students by the Instructor on Dropbox, via email or in the Boston College library.

Instructor:
Fr. Cyril P. Opeil, S.J., Ph.D., Associate Professor of Physics
Office: St. Mary's Hall, Rector's Office, 617-552-8200 or 617-552-8171.
Office: Higgins 330J, 617-552-8450; Laboratory: Higgins 130, 617-552-3589

Office Hours:
Tuesday 4:00-5:30 PM and Thursday 4:00-5:30 PM. Offices hours are only offered during these times.

To contact Fr. Opeil, S.J. outside of class use email: opeil@bc.edu or you can call my secretary, Mrs. Kelly Cunnife at kelly.cunnife@bc.edu to leave a message for me, at 617-552-8200.

Graduate Assistant:
There will be no graduate assistant this semester.

Grading Exercises, Quizzes, Presentations and Laboratory Reports:
The instructor will grade all assignments for this class.

Laboratory Reports 10 points each, 70 points total
Exercises, 35 points total
- Kaleidagraph Exercise, 5 points
- Library Exercise, 5 points
- Archimedes Exercise 1, 5 points
- Archimedes Exercise 1, 10 points
- Cp vs. T, calculating gamma & Debye temperature Exercise, 10 points

Reading quizzes, 20 points total
Student Presentation, 15 points total
- Student presentation 1, 5 points (email of presentation by due date)
- Student presentation 2, 10 points (presentation grade)

Pre-requisites: Senior/Junior Physics majors are eligible to take this class.

Course Description: PHYS4350_01 is a three-credit course designed for senior/junior Physics majors. The course consists of a series of lectures, exercises, reading assignments and laboratory exercises that focus on measurements and analysis techniques used in physics research. The experiments are designed to enable hands-on exploration of laboratory techniques. The course will be a hybrid of lecture classes (in Higgins 250) and in-class laboratory experiments (in Higgins 130).

Course Objectives: This course introduces students into experimental techniques and measurements associated with condensed-matter physics research. The techniques explored in this course are used by graduate students engaged in physics research with the intent of publication in refereed journals. Students will gain critical skills in performing laboratory techniques associated with sample making, sample preparation and temperature dependent measurements.

Course Goals:
1) Learn about the various measurement temperature dependent techniques used in condensed-matter physics research.
2) Gather and analyze experimental data utilizing computers and test equipment.
3) A series of experimental research articles will be assigned for reading and general knowledge will be tested by a quiz. The research articles are not directly related to the assigned laboratory but are used to show the breath and detail of experimental techniques.
4) One (10-12 minute) presentation will be conducted by a student on the assigned experimental technique, i.e. the Nernst Effect.

Reading Assignments/Quizzes: Scientific periodical literature is one of the major products of research and how people communicate in academics. In effort to expand the horizons of undergraduate students, scientific articles will be read throughout this course. The topics of these articles do not have a direct bearing or coordination on the experiments we will perform in this course. This is intentional. For students going on to graduate school in Physics, the experience of reading and comprehending articles will become part of your life. For students not going on to graduate school, these articles will give you an introduction to techniques beyond the time and equipment limitations of this course. A brief multiple choice question quiz will be given on each reading according to the Reading Assignments schedule each Tuesday. Copies of assigned articles are available (only to students enrolled in this class, at no cost, for educational purposes to avoid copyright infringement). Articles will arrive either by Dropbox or email.

Exercises/Homework:
Several exercises and homework assignments will be due throughout the semester. These exercises assist the student in understanding a variety of experimental and analysis techniques.

Library Exercise
Kaleidagraph/Origin Exercise
C_p analysis Exercise
Archimedes Exercise Parts 1 & 2 (Part 1 is assigned at a particular time. Part 2 can be scheduled at any time. Ask the instructor for the apparatus.)

Student Presentations on Experimental Techniques:

As a way of actively engaging questions and gaining experience about physics, experimental techniques and public presentation of your knowledge, students will give PowerPoint presentations during the last week of class. PowerPoint presentations will be given by students according to the syllabus schedule.

Each presentation will be 20-25 minutes long with 5-8 minutes for questions. Once you have chosen your topic, students are required to discuss the scope of the project with the instructor. Such a discussion normally occurs during office hours of the instructor. The deadlines for choosing a topic and for sending the completed PowerPoint presentation to the instructor are given in the syllabus schedule.

The audio presentation is required to be submitted before giving the oral presentation. See course schedule. The project submission will be a video and audio (PowerPoint, MP4-movie or Keynote) file so that the presentation can be viewed by anyone after the conclusion of the course.

Energy Project Presentations will be judged by the instructor and are assigned a grade.
+++ 0-5 points are awarded upon discussion and approval of your project with the instructor.
+++ 0-10 points are awarded for the quality of the presentation given.

In developing a talk on your topic:
History of this topic or measurement.
What underlying physics does the measurement reveal?
How is this measurement made?
Show examples of using the measurement method.
Show an example of research data from journal literature using this technique.

The criteria upon which your presentation will be judged are:
a) Description of the project undertaken.
b) Clarity of issue presented.
c) Care in presentation of the data or outcome.
d) What was learned from the project?
e) Use of research by other scientists for comparison?

Suggested topics:
Nernst Effect: what does it measure and how is it measured?
X-Ray Diffraction: what does it measure and how is it measured?
Electron Diffusion Spectroscopy: what does it measure and how is it measured?
How do you make a Neodymium Iron Boron Magnet (NdFeB) and why make them?
Magnetocaloric Cooling - What is the physics behind this type of cooling? What material candidates are effective for this type of cooling?
What is Raman Spectroscopy: what does it measure and how is it measured?

What is the Van der Pauw measurement method?
What is Hall Effect and how do we measure it?
How do vacuum pumps work? Mechanical, oil-diffusion, turbo-mechanical, ion.
The Physics behind Topological Insulators. What are they and why are they important?
Heavy Fermions - Why are these Fermions considered heavy? Why are they an important material class?
He^3 Refrigeration - How does a He^3 refrigerator work cool below the temperature of liquid He^4?
Heat capacity - How is it measured? What is physics is gained by this experiment?
Superconductivity (BCS) - How is it measured and is the physics behind this "state" of matter?
Colossal Magnetoresistance Materials - What makes them "colossal"? What is their application?
Seebeck Coefficient: what does it measure and how is it measured?
Thermal Conductivity: what does it measure and how is it measured?

Experimental Laboratory Procedures:

1. Using an ULVAC-ZEM to measure Seebeck and resistivity on Constantan and Nickel.
2. Speed of Sound in Metals at 295 K and Young's Modulus, multi-frequency probe.
3. Strain gage calibration and measurement.
4. Speed of Sound in materials at 295 K using a single frequency, through transmission.
5. Capacitance and Dielectric Measurements
6. Measuring Q in a tank circuit.
7. Measuring diamagnetic and paramagnetic materials

Laboratory Reports will be due one week after finishing the experiments in the laboratory.

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Safety: The laboratory exercises essential to in this course require that you perform them while observing appropriate safety rules, regulations and requirements. Laboratories can be dangerous places for people who do not observe safety rules, regulations and requirements. *The instructor will not tolerate students who do not observe appropriate rules, regulations and requirements. There will be no tolerance for inappropriate behavior on the part of students, because the inability to observe safety rules can result in serious permanent physical injury. If a student is asked to leave the laboratory during a laboratory experiment, for improper behavior, they will lose all points associated with that exercise.* Goggles and gloves will be required when working with materials or doing experimental work. No goggles or gloves are required to use computers in the laboratory. No one should work alone while in the laboratory. Commonsense and experience are the best defense against accidents, if you have little or no laboratory experience ask an experienced person before you perform laboratory tasks. Report any dangerous situations to the instructor immediately. Almost all laboratory accidents are preventable, the majority of accidents happen when people do not observe safety rules or do not apply their commonsense. If you have a future in experimental Physics, safety concerns will be part of your future. (Theorists have little to worry about beyond spilling their cappuccino on their keyboard.) Accidents happen, but liability issues for universities, companies and government agencies call for eliminating many problems.

Attendance, Participation and Academic Honesty: Continuity and active participation are crucial to the success of this course. You are responsible for all information from each lecture and laboratory session whether you attend or not. Absences due to (excused absence) sports activities are to be discussed with the instructor prior to the planned absence.

This course requires people to work in a group for each experimental set up. Each experimental report should be written up individually. Writing up one lab report and submitting it with different names will be graded appropriately.

Please review the Boston College Standards of Academic Integrity that are published in the university catalog and on the web at:
http://www.bc.edu/offices/stserv/academic/resources/policy/#integrity.
It is *your responsibility* to ensure that all work you submit is in accordance with university policies. If you have any questions, please consult with the instructor. Violations will be reported to the Deans' Office and reviewed by the College's Committee on Academic Integrity. Sanctions for violating this policy include the possibility of failing this course.

Class participation/Attendance: Students are expected to attend class and perform assigned laboratory exercises. Do not be afraid to ask questions. No laboratory experiments can be done on-line.

Dropbox: Assigned readings from journals, laboratory procedures, exercises, periodic announcements, the syllabus and scheduling adjustments can be found on a Dropbox link sent by the instructor to the students via email. Students will be required to sign up for a free Dropbox account. Go to "https://www.dropbox.com" on the www and just to go "sign up for free".

Laboratory Hours:
Laboratory space and equipment are available for students during the posted laboratory hours. Use your time effectively. Please, do not request personal special laboratory hours. Safety rules mandate that student laboratories must be supervised, requiring the presence of the instructor or teaching assistant. The instructor is available only during normal laboratory hours.

Disabilities: If you have a disability, please discuss the situation with the instructor. A laboratory is not a traditional classroom so accommodations to the laboratory environment may invite creative solutions. Laboratory courses require certain skills and accommodations can be made to assist students when possible and necessary.

Eating and Drinking: *Absolutely* no food or beverages are to be brought into or consumed in the lecture room or laboratory. No water bottles! Not consuming food and or beverages in the laboratory is simply a matter of safety and regulated by both state law. The laboratory is never a place to eat or drink anything; one mistake can cause

serious permanent harm or kill you. Laboratories are filled with poisonous materials, don't risk ingesting them.

Late Assignments and Laboratory Reports:
Laboratory reports must be handed in on time. A penalty of one point per day will be taken for late reports. No laboratory reports, exercises, reading quizzes or presentations will be graded for credit if received after 5 PM, on Thursday, May 11, 2023.

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List of Reading Assignments:
RA02: C. P. Opeil and M. J. Graf, "Evidence for the existence of a magnetic quantum critical point in $U(Pt_{1-x}Pd_x)_3$" *Physica B* **319**, 246-250 (2002).

RA03: C. P. Opeil, A. de Visser, M. J. Naughton, and M. J. Graf "Crossover from anomalous to conventional antiferromagnetism in Pd-doped UPt_3 studied via cantilever magnetometry" *Journal of Magnetism and Magnetic Materials* **272-276**, 244-245 (2004).

RA04: C. P. Opeil, J. C. Lashley, J. L. Smith "Angle-dependent magnetoresistance near the premartensitic phase of Ni_2MnGa" *Journal of Physics: Conference Series* **150** 042109 (2009).

RA05: C. P. Opeil, B. Mihaila, R. K. Schulze, L. Manosa, A. Planes, W. L. Hults, R. A. Fisher, P. S. Riseborough P. B. Littlewood, J. L. Smith, and J. C. Lashley "Combined experimental and theoretical investigation of the premartensitic transition in Ni_2MnGa" *Physical Review Letters* **100**, 165703 (2008).

RA06: C. P. Opeil, R. C. Albers, K. B. Blagoev, M. Gulácsi, J. C. Lashley, P. B. Littlewood, M. E. Manley, B. Mihaila, P. S. Riseborough, R. K. Schulze, D. J. Thoma, H. M. Volz and J. L. Smith "Photoelectric effect in Uranium" *Journal of the Physical Society of Japan* Vol. **75**, 56-57 (2006).

RA07: C. P. Opeil, R. K. Schulze, H. M. Volz, J. C. Lashley, M. E. Manley, W. L. Hults, R. Hanrahan, Jr., J. L. Smith, B. Mihaila, K. B. Blagoev, R. C. Albers, P. B. Littlewood "Angle-resolved photoemission and first-principles electronic structure of α-U(001)" *Physical Review B*, **75**, 045120 (2007).

RA08: C. P. Opeil, G.J. Consolmagno, D.T. Britt "The thermal conductivity of meteorites: New measurements and analysis" *Icarus* **208**, 449-454 (2010).

RA09: K. C. Lukas, W.S. Liu, Q. Jie, Z.F. Ren, C.P. Opeil, "Thermal Stability of Thermoelectric Materials via In Situ Resistivity Measurements" *Review of Scientific Instruments* **83**, 115114, 2012.

RA10: K. C. Lukas, W. S. Liu, G. Joshi, Z. F. Ren, C. P. Opeil "Thermoelectric Properties of Ho Doped $Bi_{0.88}Sb_{0.12}$" *Journal of Materials Science*, **47**, 5729-5734 (2012).

RA11: K. C. Lukas, W. S. Liu, Z. F. Ren, and C. P. Opeil "Transport properties of Ni, Co, Fe, Mn doped $Cu_{0.01}Bi_2Te_{2.7}Se_{0.3}$ for thermoelectric device applications" *Journal of Applied Physics* **112**, 054509, 2012.

RA12: H. Zhao, M. Pokharel, G. Zhu, S. Chen, K. Lukas, Q. Jie, C. Opeil, G. Chen, and Z. Ren "Dramatic thermal conductivity reduction by nanostructures for large increase in thermoelectric figure-of-merit of $FeSb_2$" *Applied Physics Letters*, **99**, 163101 (2011).

RA13: C. P. Opeil, D. T. Britt, R. J. Macke and G. J. Consolmagno, "The Surprising Thermal Properties of CM Carbonaceous Chondrites" *Meteoritics & Planetary Science* **55**, E1–E20 (2020).

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Boston College Spring 2023 Schedule

Monday, 16 January 2023 - Martin Luther King, Jr. Day - No classes
Tuesday, 17 January 2023 - Spring 2023 classes begin
Monday, 06 March - Spring vacation, no classes.
Tuesday, 07 March - Spring vacation, no classes.
Wednesday, 08 March - Spring vacation, no classes.
Thursday, 09 March - Spring vacation, no classes.
Friday, 10 March - Spring vacation, no classes.
Thursday, 06 April - Holy Thursday, no classes.
Friday, 07 April - Good Friday, no classes.
Monday, 10 April - Easter Monday, no classes.
Monday, 17 April - Patriot's Day, no classes.
Tuesday, 18 April - Substitute Monday class schedule.
Friday, 05 May - Study days, no classes.
Monday, 08 May - Study days, no classes.
Tuesday, 09 May - Term examinations begin.
Tuesday, 16 May - Term examinations end.
Monday, 22 May - Commencement.

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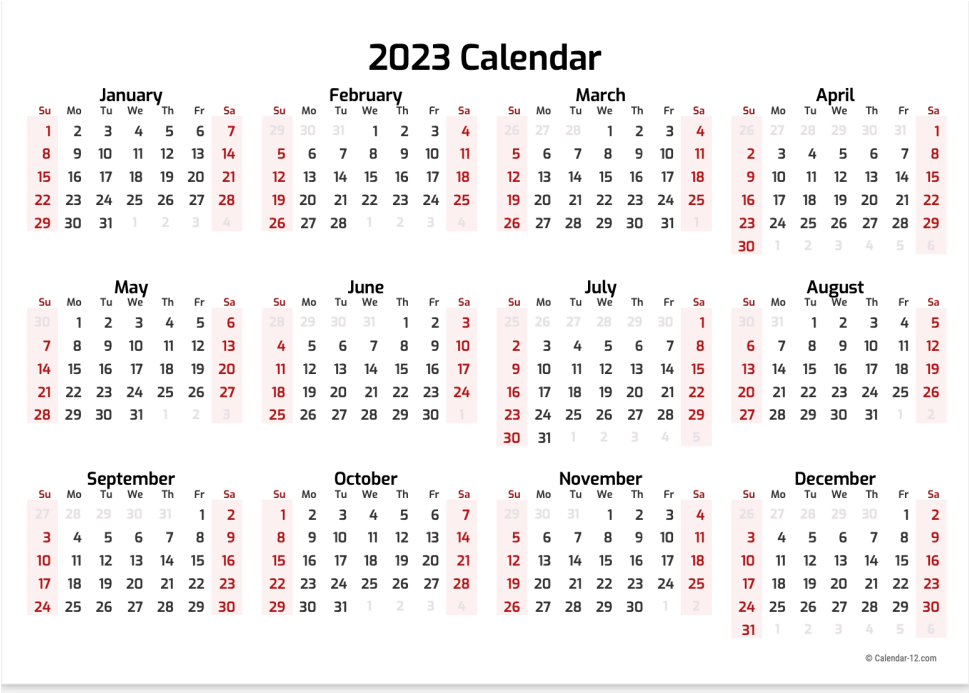

Semester Schedule: Spring 2023
Lecture & Laboratory Hours: Tuesday & Thursday 1:30-4:00 PM
Office Hours: Tuesday 4:00-5:30 PM or Thursday 4:00-5:30 PM. Offices hours are only offered during these times.

Experimental Laboratory Procedure:

1. Using an ULVAC-ZEM to measure Seebeck and resistivity on Constantan and Nickel.
2. Speed of Sound in Metals at 295 K and Young's Modulus, multi-frequency probe.
3. Strain gage calibration and measurement.
4. Speed of Sound in materials at 295 K using a single frequency, through transmission.
5. Capacitance and Dielectric Measurements
6. Measuring Q with a tank circuit.
7. Magnetism in Materials

Date	Lecture or Laboratory	Detail of Activity	Homework Assignment Exercises	Laboratory Report	Reading Quiz
Tuesday, January 17, 2023	Lecture 1, 1:30-4:00 PM Higgins 250	Classes begin: Syllabus review, Laboratory Safety review.	Exercise part 1 Archimedes Principle assigned	Laboratory reports are due one week after the end of that laboratory experiment.	N/A
Thursday, January 19, 2023	Kaleidagraph Exercise assigned during class, Higgins 250	Pre-labs for experiments and theory.	Library Exercise assigned		N/A
Tuesday, January 24, 2023	Measuring Capacitance & Dielectric Strength of Materials	Pre-labs for experiments and theory.	Exercise part 1 Archimedes Principle due at 5:00 PM		
Thursday, January 26, 2023	Measuring Capacitance & Dielectric Strength of Materials		Library Exercise due at 5:00 PM Kaleidagraph Exercise due at 5:00 PM		RA02
Tuesday, January 31, 2023	Measuring Q with an LC Circuit - the Ring-Down Method				
Thursday, February 02, 2023	Measuring Q with an LC Circuit - the Ring-Down Method			Measuring Capacitance & Dielectric Strength of Materials due at 5:00 PM	RA03

Tuesday, February 07, 2021	Magnetism in Materials				
Thursday, February 09, 2023	Magnetism in Materials		C_p Exercise assigned.	Measuring Q with an LC Circuit - the Ring-Down Method due at 5:00 PM	RA04
Tuesday, February 14, 2023	Acoustic Speed of Sound & YM in Metals				
Thursday, February 16, 2023	Acoustic Speed of Sound & YM in Metals		C_p Exercise due at 5:00 PM		RA05
Tuesday, February 21, 2023	Acoustic Speed of Sound & YM in Metals				
Thursday, February 23, 2023	Acoustic Speed of Sound & YM in Metals			Student Presentations Topics chosen	RA06
Tuesday, February 28, 2023	ZEM: Measuring Seebeck and Resistivity at 50-400 ºC			Discussion regarding the Student Presentations Topics with instructor	
Thursday, March 02, 2023	ZEM: Measuring Seebeck and Resistivity at 50-400 ºC			Acoustic Speed of Sound & YM in Metals due at 5:00 PM	RA07
Tuesday, March 14, 2023	ZEM: Measuring Seebeck and Resistivity at 50-400 ºC				
Thursday, March 16, 2023	ZEM: Measuring Seebeck and Resistivity at 50-400 ºC				RA08
Tuesday, March 21, 2023	Speed of Sound - Through Transmission				
Thursday, March 23, 2023	Speed of Sound - Through Transmission			ZEM: Measuring Seebeck and Resistivity at 50-400 ºC due at 5:00 PM	RA09
Tuesday, March 28, 2023	Speed of Sound - Through Transmission				

Thursday, March 30, 2023	Speed of Sound - Through Transmission				RA10
Tuesday, April 04, 2023	Measuring w/ Strain gages				
Tuesday, April 11, 2023	Measuring w/ Strain gages			Speed of Sound - Through Transmission due at 5:00 PM	
Thursday, April 13, 2023	Measuring w/ Strain gages			Student presentations due by email at 5:00 PM	RA12
Thursday, April 20, 2023	Measuring w/ Strain gages				RA13
Tuesday, April 25, 2023	Student presentations delivered orally in Higgins 250				
Thursday, April 27, 2023	Make up lab			Measuring w/ Strain gages due at 5:00 PM	No reading quiz
Tuesday, May 02, 2023	Make up lab		Archimedes part 2 report due by 5:00 PM.		
Thursday, May 04, 2023	Make up lab				No reading quiz

No laboratory reports, exercises, reading quizzes or presentations will be graded for credit if received after 5 PM, on Thursday, May 11, 2023.

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Boston College Policy on Sexual Harassment:
Boston College seeks to foster a campus environment that supports its educational mission and is free from exploitation and intimidation, as well as discrimination based upon gender. Sexual misconduct of any kind, including sexual harassment and sexual violence, intimate partner violence, and stalking, is antithetical to the mission of Boston College and the values it espouses and will be responded to accordingly.
The University strives to eliminate sexual violence and harassment on campus, prevent its occurrence, and address its effects. This policy describes how the University responds to sexual misconduct and how the University seeks to provide a prompt, fair and equitable response to complaints in accordance with Title IX.

This policy provides information regarding the University's education, prevention, and response efforts related to sexual misconduct by students, including descriptions of prohibited conduct, options to report misconduct (including confidential options), the process for resolving complaints, possible remedies and sanctions, and on- and off-campus resources.
The University strongly encourages any student who has been harassed or subjected to sexual misconduct to seek prompt assistance from the resources described in this policy. Further information on this topic can be found at:

https://www.bc.edu/content/dam/bc1/offices/StudentAffairs/main/pdfs/BCSSMP-2019-2020-FINAL.pdf
https://www.bc.edu/content/bc-web/offices/human-resources/sites/oid/Policies-and-Compliance/Title-IX-Boston-College.html